

5. (2.25 points)

Given the following information:

<u>Age-to-age development factors</u>					
Accident	12-24	24-36	36-48	48-60	60-72
<u>Year</u>	<u>Months</u>	<u>Months</u>	<u>Months</u>	<u>Months</u>	<u>Months</u>
2008	1.324	1.127	1.065	1.025	1.012
2009	1.313	1.127	1.058	1.027	
2010	1.344	1.135	1.070		
2011	1.340	1.134			
2012	1.344				

- $\text{Var}[Z] = 1.125$, where Z is the calendar year effect test statistic developed by Mack.
- z -value for the 95th percentile of the normal distribution: 1.645

a. (2 points)

Using the procedure shown by Mack, test the null hypothesis that the triangle does not exhibit calendar year effects at the 90% confidence level.

b. (0.25 point)

Identify one internal company action that can cause calendar year effects in a loss development triangle.

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QUESTION 5

TOTAL POINT VALUE: 2.25

LEARNING OBJECTIVE: A2: Estimate parameters and unpaid claims using claims development models related to loss reserving methods such as chain ladder, Cape Cod, chain ladder plus calendar-year effects, and Bornhuetter-Ferguson.

SAMPLE ANSWERS

Part a: 2 points

Sample Answer

	12-24	24-36	36-48	48-60	60-72
2008	<i>S</i>	<i>S</i>	<i>*</i>	<i>S</i>	<i>*</i>
2009	<i>S</i>	<i>S</i>	<i>S</i>	<i>L</i>	
2010	<i>L</i>	<i>L</i>	<i>L</i>		
2011	<i>*</i>	<i>L</i>			
2012	<i>L</i>				

Diagonal j	S_j	L_j	$N_j = \#S + \#L$	$m_j = \frac{n-1}{2}$	$Z_j = \min(\#S, \#L)$	$E(Z_j)$	$Var(Z_j)$
2	2	0	2	0	0	0.50	0.25
3	1	1	2	0	1	0.50	0.25
4	2	1	3	1	1	0.75	0.1875
5	0	4	4	1	0	1.25	0.4375
Total					2	3.00	1.125

$$E(Z_j) = \frac{n}{2} - \left(\binom{n-1}{m} \frac{n}{2^n} \right)$$

$$\text{Var}(Z_j) = \frac{n(n-1)}{4} - \left(\binom{n-1}{m} \frac{n(n-1)}{2^n} \right) + E(Z_j) - E(Z_j)^2$$

Range of Null $[3-1.645(1.125^{.5}); 3-1.645(1.125^{.5})]=(1.26, 4.74)$

Z=2 is within 90% CI, so do not reject null hypothesis that there are no calendar year effects.

No CY Effects

Part b: 0.25 point

Sample Answer 1

Company can strengthen case reserves for all AY during a particular calendar year.

Sample Answer 2

Change in the claim settlement rate, like increasing the speed of settlement starting at a given date.

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Sample Answer 3

If a company changes how it handles small claims (i.e. starts processing them faster) as of a point in time, this can affect multiple AYs and show up in the triangle as a CY effect.

Sample Answer 4

A company may change its claim processing system in a calendar year, impacting claims from all accident years.

EXAMINER'S REPORT

Candidates were expected to be able to test for calendar year impacts and identify causes of those impacts. In general, most candidates performed well on this question, with the majority of errors being caused by errors in calculations.

Part a

- Candidates were expected to be able to test whether there were calendar year impacts within a development triangle using the method outlined by Mack; "Measuring the Variability of Chain Ladder Reserve Estimates". This paper contained a numerical example illustrating the method, which candidates were expected to understand and recreate.
- Candidates generally performed well on this question, clearly demonstrating an understanding of the learning syllabus. Candidates were typically able to set up the solution correctly and errors occurred in calculating the solution. The most common errors was an error in calculating $E[Z]$ and/or not showing supporting calculations of $E[Z]$.
- Other errors included not identifying development factors as "high" or "low", miscalculating Z , or not taking the square root of Z .

Part b

- Candidates were expected to be able to identify one internal company action which may lead to a calendar year impact.
- The majority of candidates received full credit on part b of this question. Candidates who did not receive credit most commonly listed an external influence (such as inflation or legislative changes) or provided answers that were too vague, such as "reserve change".